

KinyaBERT-large

Pretrained encoder · 12 layers · 768-dim · Mean Pooling · L2 Normalize

SEQUENTIAL TRAINING - Multiple Negative Sampling Loss (MNRL)

Stage 1

Gazette Paraphrases

Kinyarwanda Legal Corpus
monolingual pairs

MNRL

sc30

scale 30

sc35

scale 35

sc40

scale 40

Stage 2

MNLI Triplets

Machine-translated rw
anchor · pos · neg

MNRL

v12

MNLI fine-tuned

Stage 3

OPUS-100 en↔rw

Cross-lingual parallel
English ↔ Kinyarwanda

MNRL

step22A

OPUS cross-lingual checkpoint

Stage 4

KinyaCOMET

2,936 human pairs (score ≥ 0.8)
Novel en-rw annotation

MNRL

step23A

KinyaCOMET checkpoint · $\times 2$ weight

$\times 2$ weight in ensemble

avg

Ensemble

all5 + step23A $\times 2$

sc30

sc35

sc40

v12

step22A

step23A $\times 2$

Sum 7 embeddings
Divide by 7
L2 Normalize

768-dim Sentence Embedding

Cosine-ready · Kinyarwanda + cross-lingual